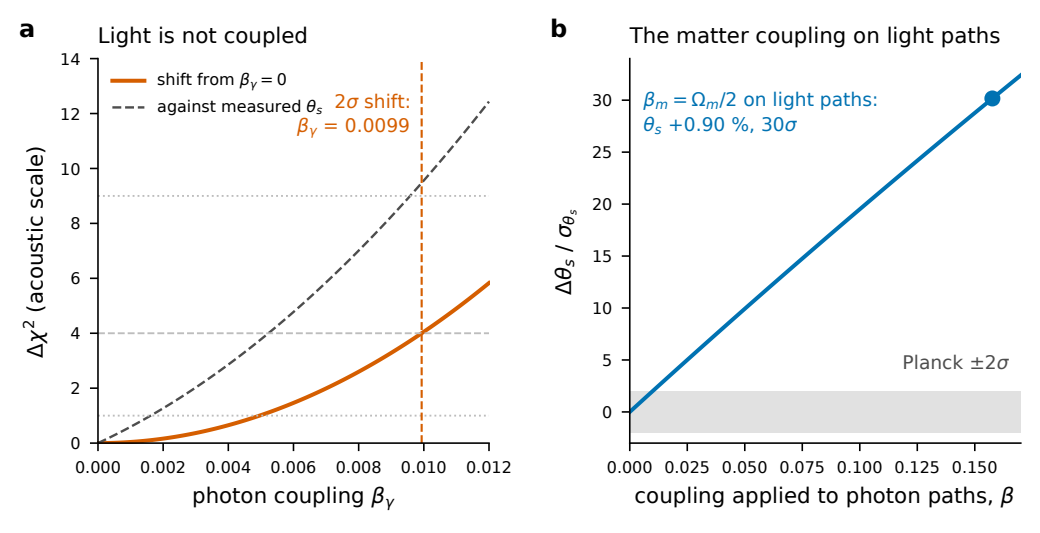
a
Light is not coupled
shift from $\beta_\gamma = 0$
against measured θ_s
2σ shift:
$\beta_\gamma = 0.0099$
$\Delta\chi^2$ (acoustic scale)
photon coupling β_γ
b
The matter coupling on light paths
$\beta_m = \Omega_m/2$ on light paths:
θ_s +0.90 %, 30σ
Planck $\pm 2\sigma$
$\Delta\theta_s$ / σ_{θ_s}
coupling applied to photon paths, β